Supplementary Material for "Knowing when to ask a human: reliable confidence for fast time series classifiers under sensor faults"

Author information removed for double-anonymous review

See separate title page.

Abstract

Proofs of Propositions 1–2 of the main text, the deviations log, the leakage audit, the definitions of the simulated faults, the validation of the software tool, and additional tables and figures. All numbers are generated from the released result files.

Keywords: Supplementary material

S1 Proofs

Notation follows Sect. 3 of the main text: $A = \tilde{X}^\top \tilde{X} + \mathrm{diag}(0, \lambda I_p)$, $B = A^{-1}\tilde{X}^\top Y$, $\hat{Y} = \tilde{X}B$, $d_i = \tilde{x}_i^\top A^{-1}\tilde{x}_i$, and $A_{-i} = A - \tilde{x}_i\tilde{x}_i^\top$ is the matrix without example i (same λ and standardisation, Assumption A4). The Sherman–Morrison identity gives

$$A_{-i}^{-1} = A^{-1} + \frac{A^{-1}\tilde{x}_i\tilde{x}_i^\top A^{-1}}{1 - d_i}, \qquad \tilde{x}_i^\top A_{-i}^{-1} = \frac{\tilde{x}_i^\top A^{-1}}{1 - d_i}. \tag{S1}$$

Proposition 1 (Proposition 1 of the main text) *$z_{-i} = Y_i - (Y_i - \hat{Y}_i)/(1 - d_i)$ and $h_{-i} = d_i/(1 - d_i)$.*

Proof $B_{-i} = A_{-i}^{-1}(\tilde{X}^\top Y - \tilde{x}_i Y_i)$, so by (S1), $\tilde{x}_i^\top B_{-i} = (\hat{Y}_i - d_i Y_i)/(1 - d_i) = Y_i - (Y_i - \hat{Y}_i)/(1 - d_i)$. This is the classical PRESS identity (Allen 1974). For the leverage, $\tilde{x}_i^\top A_{-i}^{-1}\tilde{x}_i = d_i + d_i^2/(1 - d_i) = d_i/(1 - d_i)$. □

Proposition 2 (closed forms from one thin SVD; Eq. (2) of the main text) *With* $\Phi_c = USV^\top$ *and* $D_\lambda = \mathrm{diag}(1/(s_j^2+\lambda))$*:* $d_i = 1/n + \sum_j u_{ij}^2 s_j^2/(s_j^2+\lambda)$*,* $h(x) = 1/n + \|D_\lambda^{1/2} V^\top \psi\|^2 + (\|\psi\|^2 - \|V^\top\psi\|^2)/\lambda$ *and* $k(x,i) = 1/n + (V^\top\psi)^\top D_\lambda S u_i$*.*

Proof Because Φ_c is centred, the intercept column is orthogonal to the features and A is block diagonal; the intercept contributes $1/n$. For the feature block, $(\Phi_c^\top\Phi_c + \lambda I)^{-1} = VD_\lambda V^\top + \lambda^{-1}(I - VV^\top)$. Training points lie in the span of V, which gives d_i and $k(x,i)$; a test point may not, which gives the last term of $h(x)$. □

Proposition 3 (Proposition 2 of the main text) *With residuals* $E = Y - \hat{Y}$*,* $z_{-i}(x) = z(x) - k(x,i)E_i/(1-d_i)$ *and* $h_{-i}(x) = h(x) + k(x,i)^2/(1-d_i)$*.*

Proof From (S1), $B_{-i} = B - A^{-1}\tilde{x}_i Y_i + A^{-1}\tilde{x}_i(\hat{Y}_i - d_i Y_i)/(1-d_i) = B - A^{-1}\tilde{x}_i E_i/(1-d_i)$; multiplying by $\tilde{x}(x)^\top$ gives the first identity. The second is (S1) applied to $\tilde{x}(x)^\top A_{-i}^{-1}\tilde{x}(x)$. □

Under the Bayesian linear model with prior $\beta \sim \mathcal{N}(0, \sigma^2\lambda^{-1}I)$ and noise variance σ^2 (flat intercept), the predictive variance of each score is $\sigma^2 h(x)$, so $z(x)/\sqrt{h(x)}$ is the predictive mean divided by its standard deviation up to σ. All identities are checked against explicit refits by unit tests in the code repository. Results 1 and 3 are standard Sherman–Morrison algebra; we state them in the form needed here.

S2 Deviations log

Every change to the registered protocol, with date, reason, timing relative to the results and effect. Entries marked post hoc were made after results were known.

ID	Date	Change	Reason	Timing	Effect
D1	2026-10-13	Platt target smoothing added to the LOOM fit	GunPoint smoke test: separable LOO predictions drove the temperature to its bound	before the benchmark	Prevents infinite temperatures; raises clean ECE on few-shot datasets (reported)
D2	2026-10-13	APS score added to the conformal analysis	Quick-mode run on 6 development datasets: LAC sets empty under shift	before the benchmark	Adds a set score; LAC still reported
D3	2026-10-13	Ensembles only on the official split	One job took 68 CPU-min; container restarts	before results (4 finished jobs discarded)	Ensembles compared on 114 datasets, official split only
D4	2026-10-13	Conformal sets on $\leq$ 1,000 seeded test series	compute	before results	Coverage estimated on a subset for large test sets
D5	2026-10-13	Rocket dropped from E5; E5 and E7 use clean + severity 3 only	compute (Rocket transform $\sim$10$\times$ MiniRocket)	before results	Generalisation shown for MultiRocket and Hydra only
D6	2026-10-20	Crop and ElectricDevices test sets subsampled to 2,000 series	leverage projection $\sim 10^{13}$ operations	before results of these datasets	Same subsample for all methods

ID	Date	Change	Reason	Timing	Effect
D7	2026-10-20	FordA, FordB, ElectricDevices: resamples 0–2 only; E5/E7 exclude the four largest datasets	out-of-memory kills, ∼16 h remaining	before results of these datasets	3 instead of 5 resamples for three datasets
D8	2026-10-13	LogReg-CV skipped for > 2,000 training series; Arsenal in MiniRocket-kernel configuration, official split, clean + severity 3	compute	before the benchmark	LogReg n = 110 (109 with failures), Arsenal n = 113
D9	2026-11-02	Arsenal returned NaN probabilities on Fungi (one series per class); CV calibrators undefined for classes with < 5 series	method limitations	after results	Those methods compared on fewer datasets (n reported)
D10	2026-10-21	Official-split table (E2) restricted to severity-3 conditions for every method	an earlier version mixed condition sets, favouring LOOM against the Arsenal	post hoc	QUANT becomes significantly better than LOOM there; reported
D11	2026-10-21	Natural cross-axis shift experiment (E10) added	reviewer-style objection that faults were only synthetic	post hoc	LOOM's gain is small there; reported as a limitation
D12	2026-10-21	LogReg convergence check (E16)	suspicion of an under-trained baseline	post hoc	Median NLL unchanged; one many-class dataset improves by 0.16; reported
D13	2026-10-29	Exploratory triage study (E11) designed after the benchmark; protocol committed before running	practical framing of the paper	post hoc design	Reported as exploratory; confirmed later by E13
D14	2026-10-21	Ridge-onehot/softmax fit times include the shared SVD; MiniRocket transform time reported separately	accounting error found in review	post hoc (bug fix)	Table of costs corrected; no benchmark rerun
D15	2026-11-02	E13 dataset count corrected from 39 to 38	counting error in the registration text	before running E13	inclusion rule unchanged
D16	2026-11-02	NSTDB records read with an own format-212 reader instead of the `wfdb` package	`wfdb` incompatible with pandas 3	before running E12	none; reader checked against header initial values
D17	2026-11-02	E12 noise windows taken at the native 360 Hz sampling of NSTDB (no resampling to the unknown UCR sampling rates)	UCR files do not record sampling rates	written into the E12 registration	Noise bandwidth relative to the ECG beats is approximate
D18	2026-11-03	Numeric pass/fail criteria for H1–H3 written down	protocol template requires explicit criteria	post hoc (wording)	Criteria restate the registered tests; H1b fails under them
D19	2026-11-03	Sanity checks E17 (sampling-noise simulation, exchangeability check) and planted-leak test added	protocol template	post hoc	Diagnostics only; no change to the main analysis
D20	2026-11-03	Means with 95 % bootstrap confidence intervals reported alongside the registered medians and Wilcoxon tests	protocol template	post hoc (reporting)	No change to the registered tests
D21	2026-11-04	Tool validation (E18): `loom-triage` run on plain CSV files built from the raw UCR text files, compared with the pipeline	protocol template requires validating the tool outside the pipeline	post hoc	Identical predictions on 5 of 5 datasets; confidences equal to machine precision

S3 Leakage audit

Exact duplicates (rounded to six decimals) and near-duplicates (z-normalised RMS distance to the nearest training series below 0.01 standard deviations) between the official training and test sets were searched in all 114 datasets (Table S2). 16 datasets contain exact duplicates and 24 exact or near duplicates. In ElectricDevices, many duplicated test series carry a different label from their identical training series, a data-quality issue of the archive. *Planted-leak test*: five test series of GunPoint were copied into its training set exactly and five with noise of 0.001 standard deviations; the audit flagged exactly these and nothing else (passed). Refitting LOOM after permuting the test labels gave identical parameters and predictions, confirming that the test set is never used for fitting. Results without the affected datasets are in Sect. 6.10 of the main text.

Table S2 Datasets with exact or near train/test duplicates in the official splits.

Dataset	Train	Test	Exact dup.	Near dup.	Label conflicts
ACSF1	100	100	0	12	0
Adiac	390	391	0	12	0
ChlorineConcentration	467	3840	0	36	0
Crop	7200	16800	2753	2753	0
ElectricDevices	8926	7711	208	218	101
FaceAll	560	1690	1	1	0
FreezerRegularTrain	150	2850	1	1	0
FreezerSmallTrain	28	2850	27	27	0
InlineSkate	100	550	6	6	0
InsectEPGSmallTrain	17	249	15	15	0
Meat	60	60	0	40	0
MedicalImages	381	760	1	4	0
MixedShapesRegularTrain	500	2425	1	1	0
OliveOil	30	30	0	30	0
PowerCons	180	180	67	67	0
ShapesAll	600	600	2	2	0
SmallKitchenAppliances	375	375	0	1	0
Strawberry	613	370	1	48	1
SwedishLeaf	500	625	2	2	0
ToeSegmentation1	40	228	4	4	0
Wafer	1000	6164	0	244	0
Wine	57	54	0	38	0
Worms	181	77	10	10	0
WormsTwoClass	181	77	10	10	0

S4 Simulated faults

Faults are applied to test series only, with amplitudes relative to each series' standard deviation sd: noise (Gaussian, $\sigma \in \{0.1, 0.3, 0.5\}$ sd); baseline wander (sinusoid of amplitude $\{0.25, 0.5, 1\}$ sd, 0.5–2 cycles per series); dropout (a segment of $\{10, 20, 30\}$ % replaced by linear interpolation); spikes (± 3 sd at $\{1, 3, 5\}$ % of time points); speed variation (smooth monotone time warp with local slope in $[1-s, 1+s]$,

$s \in \{0.05, 0.1, 0.2\}$); misalignment (circular shift of $\{5, 10, 20\}$ % of the length). Random draws are seeded by dataset, resample, fault and severity. In the triage studies, cardiac and industrial datasets receive noise, wander, spikes or dropout, and motion datasets noise, spikes, dropout or speed variation. Real noise (MIT-BIH NSTDB records bw, em and ma, both channels) is added as a random window of the series length at its native 360 Hz sampling, scaled to 12, 6 or 0 dB SNR relative to each series' variance.

S5 Validation of the software tool

The command-line tool was run as a separate process on plain CSV files built directly from the official UCR text files of five datasets (ECG200, GunPoint, ItalyPowerDemand, TwoLeadECG, Wafer), with `fit`, `run` (20 % review budget, HTML report) and `evaluate`. Its predictions were identical to those of the research pipeline on five of five datasets, and confidences differed by at most 1.1×10^{-16}. The package also has unit tests for the API and the command-line interface.

S6 Additional tables and figures

Table S3 Prediction sets at $\alpha = 0.1$: mean over datasets (faults: over datasets and 18 conditions) of coverage, set size and share of empty sets.

	Clean			Faults		
Method	Coverage	Size	Empty	Coverage	Size	Empty
LOOM, jackknife+, LAC	0.898	1.55	4.6 %	0.638	1.66	25.0 %
LOOM, jackknife, LAC	0.899	1.51	4.4 %	0.639	1.62	24.8 %
LOOM, jackknife+, APS	0.908	5.22	7.3 %	0.905	6.62	1.5 %
PreVal, jackknife+, LAC	0.908	1.58	3.3 %	0.696	1.55	9.3 %
PreVal, jackknife+, APS	0.923	5.24	5.8 %	0.817	5.03	10.1 %
Split conformal, LAC	0.928	2.49	2.2 %	0.763	2.38	6.8 %
Split conformal, APS	0.948	5.86	3.5 %	0.876	5.41	5.8 %

References

Allen DM (1974) The relationship between variable selection and data agumentation and a method for prediction. Technometrics 16(1):125–127. https://doi.org/10.1080/00401706.1974.10489157

Table S4 Fit and prediction time per dataset (single CPU core; mean over resamples, then median or mean over datasets). Readouts include the ridge SVD but not the shared MiniRocket transform, listed separately.

Method	Median fit (s)	Mean fit (s)	Median predict (s)
MiniRocket transform (shared)	0.168	1.14	–
Ridge one-hot	0.067	3.33	0.09
PreVal	0.144	3.63	0.10
LOOM (ours)	0.140	3.61	1.49
Hold-out TS	0.053	2.03	0.30
Ridge + Platt (5-fold)	0.256	12.78	0.36
Logistic regression (CV)	3.666	11.59	0.40
LOOM, 5-seed ens.	2.247	24.93	26.26
QUANT	1.298	5.72	5.56
Arsenal	4.475	22.98	20.45

Table S5 Ablation on clean data (median NLL and ECE; W/D/L and Holm-adjusted p for NLL against LOOM).

Variant	NLL	ECE	W/D/L	p
LOOM-free-b	0.280	0.066	30/0/84	$p < 0.001$
LOOM-finite-c1	0.287	0.070	25/0/89	$p < 0.001$
PreVal-smooth	0.290	0.065	39/0/75	$p = 0.038$
LOOM-kNN	0.292	0.075	58/0/56	$p = 1.000$
LOOM	0.295	0.075	–	–
LOOM-lambdaSSE	0.300	0.076	49/26/39	$p = 0.079$
LOOM-nosmooth	0.348	0.041	53/1/60	$p = 0.541$
PreVal	0.360	0.040	49/0/65	$p = 1.000$
LOOM-holdout	0.441	0.136	110/0/3	$p < 0.001$
LOOM-insample	0.978	0.478	113/0/1	$p < 0.001$

Table S6 Ablation under simulated faults (median NLL over 18 conditions and detection AUROC).

Variant	NLL	AUROC	W/D/L	p
LOOM	0.738	0.765	–	–
LOOM-lambdaSSE	0.751	0.775	42/26/46	$p = 1.000$
LOOM-kNN	0.771	0.725	52/0/62	$p = 1.000$
LOOM-finite-c1	0.775	0.729	50/0/64	$p = 1.000$
LOOM-holdout	0.846	0.754	94/0/19	$p < 0.001$
LOOM-free-b	0.964	0.653	80/0/34	$p < 0.001$
LOOM-nosmooth	1.049	0.747	71/1/42	$p = 0.003$
LOOM-insample	1.096	0.770	108/0/6	$p < 0.001$
PreVal-smooth	1.172	0.553	94/0/20	$p < 0.001$
PreVal	1.728	0.563	108/0/6	$p < 0.001$

Table S7 Training-set size: 25 % and 50 % of the official training sets; median NLL on clean data and under severity-3 faults.

Fraction	Data	Competitor	n	LOOM	Competitor NLL	W/D/L	p
25 %	clean	PreVal	111	0.542	0.601	45/0/66	$p = 0.094$
25 %	clean	Ridge-softmax	111	0.542	0.677	92/0/19	$p < 0.001$
25 %	clean	Holdout-TS	111	0.542	0.880	82/0/29	$p < 0.001$
25 %	shifted	PreVal	111	1.058	2.155	97/0/14	$p < 0.001$
25 %	shifted	Ridge-softmax	111	1.058	1.474	91/0/20	$p < 0.001$
25 %	shifted	Holdout-TS	111	1.058	3.079	105/0/6	$p < 0.001$
50 %	clean	PreVal	111	0.442	0.477	41/0/70	$p = 0.020$
50 %	clean	Ridge-softmax	111	0.442	0.633	97/0/14	$p < 0.001$
50 %	clean	Holdout-TS	111	0.442	0.631	79/0/32	$p < 0.001$
50 %	shifted	PreVal	111	1.049	2.543	103/0/8	$p < 0.001$
50 %	shifted	Ridge-softmax	111	1.049	1.447	93/0/18	$p < 0.001$
50 %	shifted	Holdout-TS	111	1.049	2.609	107/0/4	$p < 0.001$
100 %	clean	PreVal	111	0.308	0.332	35/0/76	$p = 0.003$
100 %	clean	Ridge-softmax	111	0.308	0.574	103/0/8	$p < 0.001$
100 %	clean	Holdout-TS	111	0.308	0.475	69/0/42	$p = 0.003$
100 %	shifted	PreVal	111	1.014	2.354	102/0/9	$p < 0.001$
100 %	shifted	Ridge-softmax	111	1.014	1.302	94/0/17	$p < 0.001$
100 %	shifted	Holdout-TS	111	1.014	3.067	100/0/11	$p < 0.001$

Table S8 Other feature extractors (official split): LOOM against each readout in NLL.

Features	Data	Competitor	W/D/L	p
multirocket	clean	PreVal	38/0/72	$p = 0.008$
multirocket	clean	Ridge-softmax	100/0/10	$p < 0.001$
multirocket	clean	Holdout-TS	70/0/40	$p < 0.001$
multirocket	severity 3	PreVal	105/0/5	$p < 0.001$
multirocket	severity 3	Ridge-softmax	102/0/8	$p < 0.001$
multirocket	severity 3	Holdout-TS	107/0/3	$p < 0.001$
hydra	clean	PreVal	43/0/67	$p = 0.147$
hydra	clean	Ridge-softmax	102/0/8	$p < 0.001$
hydra	clean	Holdout-TS	74/0/36	$p < 0.001$
hydra	severity 3	PreVal	106/0/4	$p < 0.001$
hydra	severity 3	Ridge-softmax	99/0/11	$p < 0.001$
hydra	severity 3	Holdout-TS	107/0/3	$p < 0.001$

Table S9 Natural cross-axis change on UWaveGestureLibrary: means over six axis pairs and five resamples.

Method	NLL in-domain	NLL other axis	Acc. other axis	AUROC
Holdout-TS	0.595	2.548	0.322	0.708
LOOM	0.579	2.469	0.323	0.727
LogReg-CV	0.521	2.965	0.345	0.674
PreVal	0.573	2.550	0.323	0.712
Ridge-Platt-CV5	0.544	2.470	0.309	0.697
Ridge-softmax	1.187	1.894	0.322	0.718

Table S10 Convergence check of the logistic-regression baseline (12 random datasets): test NLL with default (300 iterations, tolerance 10^{-3}) and long training (3,000 iterations, 10^{-5}), and the selected regularisation C.

Dataset	NLL default	NLL long	C default	C long
ArrowHead	0.427	0.434	0.01	0.01
Car	0.348	0.345	0.01	0.01
DiatomSizeReduction	0.393	0.394	0.001	0.001
GunPointAgeSpan	0.070	0.069	0.01	0.01
GunPointMaleVersusFemale	0.008	0.001	0.1	1.0
GunPointOldVersusYoung	0.000	0.000	1.0	1.0
InsectWingbeatSound	0.933	0.933	0.001	0.001
OSULeaf	0.194	0.197	0.01	0.01
Phoneme	2.583	2.584	0.001	0.001
ShapesAll	0.599	0.437	0.1	1.0
SmoothSubspace	0.168	0.138	0.01	1.0
Strawberry	0.074	0.076	0.1	0.1

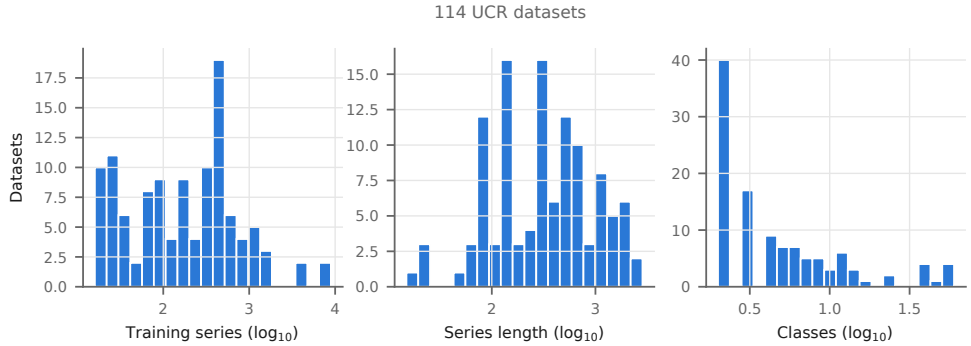


Fig. S1 Training-set size, series length and number of classes of the 114 UCR datasets.

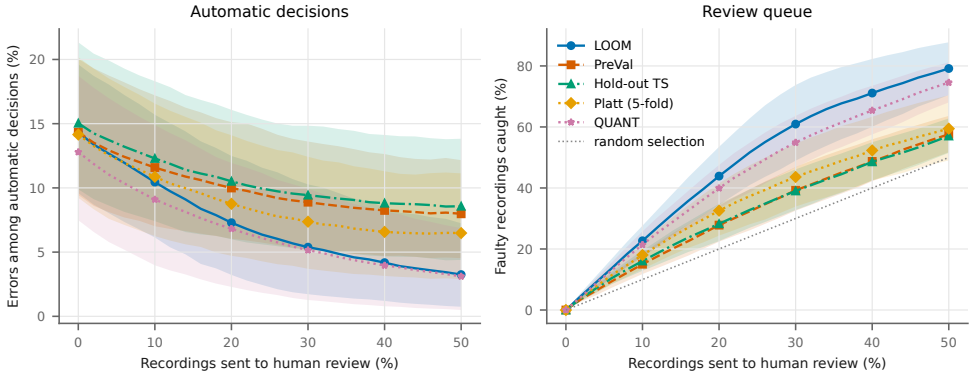


Fig. S2 Exploratory triage study (12 datasets), same layout as Fig. 6 of the main text.

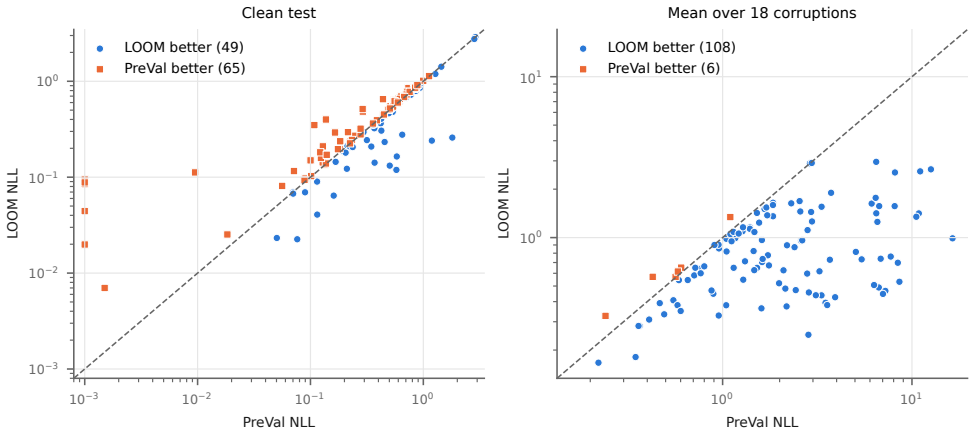


Fig. S3 Per-dataset NLL of LOOM against PreVal on clean data (left) and averaged over 18 simulated faults (right); points below the diagonal favour LOOM; values below 10^{-3} are drawn at 10^{-3}.

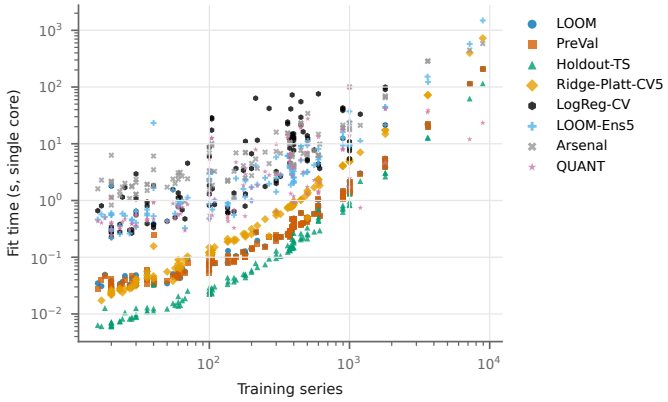


Fig. S4 Fit time against the number of training series (log–log, single core).